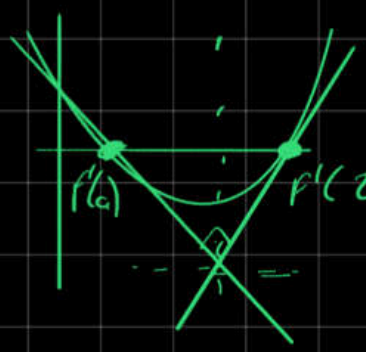
1) $y = 4(x-a)(x-2a)$

$y = (4x-4a)(x-2a)$

$= 4x^2 - 8xa - 4xa + 8a^2$

$= 4x^2 - 12xa + 8a^2$



$a \neq$

$y' = 8x - 12a$ $\quad y'(0) = -4a$ $\quad y'(2a) = 4a$

Max - Min

Asimptot

Grafik (Paydalı fonk)

Kısmi İntegrasyon

$-4a \cdot 4a = -1$

$-16a = -1$

$a = \frac{1}{16}$

ca yalış

2) $x^2 - 6xy - a \cdot y^2 = b$ $\quad (0,1)$ teğetim denk $\Rightarrow x + y = 1$ $\quad a, b$?

$y' = -\frac{2x - 6y}{-6x - 2y \cdot a} = \frac{2x - 6y}{6x + 2y \cdot a}$ $\quad y'(0,1) = \frac{-6}{2a}$ $\quad (0,1)$

$x - \frac{3x}{a} + \not{x} = \not{1}$

$y - 1 = \left(\frac{-3}{a}\right) x$

$x = \frac{3x}{a}$ $\quad a = 3$ $\quad b = -3$

$y = -\frac{3x}{a} + 1$

3) $e^y + xy - e = 0 \longrightarrow y' = -\frac{y}{e^y + x}$ $\quad \frac{\text{Mx}}{-1}$ $\quad \frac{\text{noktalı}}{e}$ $\quad (0,1)$

$x = 0 \rightarrow e^y - e = 0$ $\quad (0,1) \longrightarrow$

$e^y = e$

$y = 1$

$e^y \not{x}$

$y - 1 = \frac{-x}{e}$

$y = -\frac{x}{e} + 1$

4) $x^2y^3 + 2y + x^3 - 6 - x = 0$ $x=0$ tegel d?

$y' = -\frac{2xy^3 + 3x^2 - 1}{3y^2x^2 + 2}$ $m_t = -\frac{-1}{2} = \frac{1}{2}$, (0,3)

$x=0 \rightarrow 2y = 6$ $y=3$ $(0,3)$ $y-3 = \frac{x}{2}$

$y = \frac{x}{2} + 3$

5) $x^2 - xy + y^2 - 7 = 0$ $(-1,2)$ tegel d?

$y' = -\left(\frac{2x - y}{-x + 2y}\right)$ $y'(-1,2) = m_t = -\left(\frac{-2-2}{1+4}\right) = \frac{4}{5}$, (-1,2)

$y = \frac{4x + 14}{5}$

6) $x^3y^3 + y^2 = x + y$ $(1,1)$, $(1,-1)$ teg. egy.?

$x^3y^3 + y^2 - x - y = 0 \rightarrow y' = -\left(\frac{3x^2y^3 - 1}{3y^2x^3 + 2y - 1}\right)$

$(1,1) \rightarrow y' = -\left(\frac{3-1}{3+2-1}\right) = -\frac{2}{4} = -\frac{1}{2}$

$(1,-1) \rightarrow y' = -\left(\frac{-3-1}{3-2-1}\right) = \frac{4}{0} = \infty$?

7) $\left.\begin{matrix} x = t^2 + t \\ y = \cos 2t \end{matrix}\right\}$ Fenk $(0,1)$ tegel d?

$$y' = \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dt} = -\sin 2t \cdot 2 \qquad \frac{dx}{dt} = 2t+1 \qquad \frac{dt}{dx} = \frac{1}{2t+1}$$

$$y' = \frac{-\sin 2t \cdot 2}{2t+1} \xrightarrow{t=0} \frac{0}{1} = 0 = m_T \quad (0,1)$$

$$y-1 = 0 \cdot x$$

$$y = 1$$

g) $y = \frac{x}{2} + \frac{1}{2x-4}$ $\quad m_{teğet} = -\frac{3}{2}$ old. noktaları?

$$y = \frac{x}{2} + (2x-4)^{-1}$$

$$y' = \frac{1}{2} + -1(2x-4)^{-2} \cdot 2$$

$$= \frac{1}{2} - \frac{2}{(2x-4)^2} = -\frac{3}{2}$$

$$\frac{2}{(2x-4)^2} = \frac{1}{2} - \left(-\frac{3}{2}\right) = 2$$

$$(2x-4)^2 = 1$$

$2x-4 = 1$ $\quad \vee$ $\quad 2x-4 = -1$

$2x = 5$ $\quad\quad$ $2x = 3$

$x_1 = \frac{5}{2}$ $\quad\quad$ $x_2 = \frac{3}{2}$

$y_1 = \frac{9}{4}$ $\quad\quad$ $y_2 = -\frac{1}{4}$

$\left(\frac{5}{2}, \frac{9}{4}\right)$ $\quad\quad$ $\left(\frac{3}{2}, -\frac{1}{4}\right)$

9) $y^2 = 20x$ x değeri ile yanı ile 45° açı teğetin denklemi?

$y = (20x)^{1/2}$ $\quad m_T = 1$

$y' = \frac{1}{2}(20x)^{-\frac{1}{2}} \cdot 20 = \frac{10}{\sqrt{20x}} = 1$ $\quad \sqrt{20x} = 10$

$20x = 100$

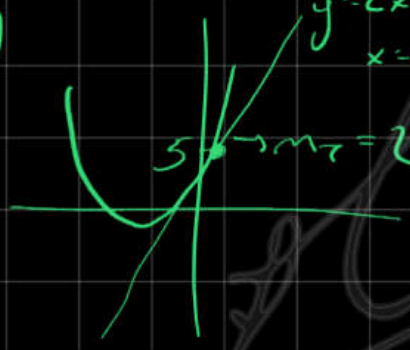
$x = 5$ $\quad y = \pm 10$

$m_T = 1, \ (5, 10)$

$y - 10 = 1(x - 5)$

$y = x + 5$

10) $f(x) + f'(x)$



$f(0) + f'(0) = 5$

$-f'(0) + f''(0) = 2$

$f(0) - f''(0) = 3$

11) $\lim_{x \to 0^+} (\ln \cot x)^{\tan x} = ?$ $\quad \infty^0$ bls.

$y = (\ln \cot x)^{\tan x} \rightarrow \ln y = \ln (\ln \cot)^{\tan x} = \tan x \cdot \ln(\ln \cot x)$

$\lim_{x \to 0^+} \tan x \cdot \ln(\ln \cot x) = 0 \cdot \infty$ bls. $\rightarrow \lim_{x \to 0^+} \frac{\ln(\ln \cot x)}{\cot x} = \frac{\infty}{\infty}$ bls.

$\lim_{x \to 0^+} \frac{\frac{-\operatorname{cosec}^2 x}{\ln \cot x}}{-\operatorname{cosec}^2 x} = \lim_{x \to 0^+} \frac{1}{\ln \cot x} = 0$

$\ln y = 0 = \log_e y$

$e^0 = y = 1$

12) $\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} = 1^{\infty}$ bls

$y = \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$ $\quad \ln y = \frac{1}{x^2} \cdot \ln\left(\frac{\sin x}{x}\right)$

$$\lim_{x\to 0} \frac{\ln\left(\frac{\sin x}{x}\right)}{x^2} = \frac{0}{0} \xrightarrow{l'} \frac{\left(\frac{x\cdot\cos x - \sin x}{x^2}\right)}{2x}$$

$$= \frac{\left(\frac{x\cdot\cos x - \sin x}{x\cdot\sin x}\right)}{2x} = \frac{\left(\cot x - \frac{1}{x}\right)}{2x} = \frac{\left(\frac{\cos x}{\sin x} - \frac{1}{x}\right)}{2x} = \frac{x\cdot\cos x - \sin x}{2x^2\cdot\sin x}$$

$$\lim_{x\to 0} \frac{x\cdot\cos x - \sin x}{2x^2\cdot\sin x} = \frac{0}{0} \xrightarrow{l'} \frac{\cos x - x\sin x - \cos x}{4x\sin x + 2x^2\cos x} = \frac{-x\sin x}{2x(2\sin x + x\cos x)}$$

$$\lim_{x\to 0} \frac{-\sin x}{2(2\sin x + x\cdot\cos x)} = \lim_{x\to 0} \frac{-\frac{\sin x}{x}}{4\frac{\sin x}{x} + 2\cos x} = \frac{-1}{6}$$

$\ln y = \frac{-1}{6}$ $\quad \log_e y = \frac{-1}{6}$

$e^{-\frac{1}{6}} = y$

43) $\lim_{x\to\infty} (e^x + e^{2x})^{1/x} = \infty^0$ bls

$y = (e^x + e^{2x})^{1/x}$ $\quad \ln y = \frac{1}{x} \cdot \ln(e^x) \cdot \ln(e^{2x})$

$\lim_{x\to\infty} \frac{\ln(e^x + e^{2x})}{x} = \frac{\infty}{\infty} \longrightarrow \frac{e^x + 2e^{2x}}{e^x + e^{2x}}$

$\lim_{x\to\infty} \left(\frac{e^x + 2e^{2x}}{e^x + e^{2x}}\right) = \lim_{x\to\infty} \frac{e^{2x}\left(\frac{1}{e^{2x}} + 2\right)}{e^{2x}\left(\frac{1}{e^{2x}} + 1\right)} = 2$

$\ln y = 2 = \log_e y$ $\quad e^2 = y$

44) $\lim_{x\to\infty} \left(1 - \sin\frac{1}{2x}\right)^{6x} = 1^\infty$

$y = \left(1 - \sin\frac{1}{2x}\right)^{6x} \xrightarrow{\ln} \ln y = 6x \cdot \ln\left(1 - \sin\frac{1}{2x}\right)$

$\lim_{x\to\infty} 6x \cdot \ln\left(1 - \sin\frac{1}{2x}\right) = \infty \cdot 0 \longrightarrow \frac{\ln\left(1 - \sin\frac{1}{2x}\right)}{\frac{1}{6x}} = \frac{0}{0}$

tru

$\xrightarrow{'} \frac{\left(\frac{-\cos\frac{1}{2x} \cdot \frac{-1}{2x^2}}{1 - \sin\frac{1}{2x}}\right)}{\frac{-1}{6x^2}} = \frac{-\cos\frac{1}{2x} \cdot \frac{-1}{2x^2}}{1 - \sin\frac{1}{2x}} \cdot \frac{-6x^2}{1} = -\frac{3\cos\frac{1}{2x}}{1 - \sin\frac{1}{2x}}$

$= \lim_{x\to\infty} \left(\frac{-3\cos\frac{1}{2x}}{1 - \sin\frac{1}{2x}}\right) = \frac{-3}{1}$ $\quad \ln y = -3$ $\quad e^{-3} = y$

15) $\lim_{x\to 0} (\cos x)^{1/x^2} = 1^{\infty}$ bls

ca x

$y = (\cos x)^{\frac{1}{x^2}}$ $\quad \ln y = \dfrac{\ln(\cos x)}{x^2}$

$\lim_{x\to 0} \dfrac{\ln(\cos x)}{x^2} = \dfrac{0}{0} \longrightarrow \dfrac{-\tan x}{2x} = \dfrac{0}{0} \longleftrightarrow \lim_{x\to 0} \dfrac{-1}{2}\dfrac{\tan x}{x}$

$\ln y = -\dfrac{1}{2}$ $\quad e^{-\frac{1}{2}} = y$

16) $\lim_{x\to \frac{\pi}{2}} (\pi \sec x - 2x\tan x) = \infty - \infty$ bls

$\lim_{x\to \frac{\pi}{2}} \left(\dfrac{\pi - 2x\cdot \sin x}{\cos x} \right) = \dfrac{0}{0} \longrightarrow \dfrac{-2\cdot \sin x + \cos x\, 2x}{-\sin x} = \dfrac{-2}{-1}$

$= 2$

17) $\lim_{x\to -\frac{\pi}{2}} (\tan x)^{\cos x} = -\infty^{0}$ bls

$y = (\tan x)^{\cos x}$ $\quad \ln y = \cos x \cdot \ln(\tan x)$

$\lim_{x\to -\frac{\pi}{2}} \cos x \cdot \ln(\tan x) = 0\cdot\infty \longrightarrow \lim_{x\to -\frac{\pi}{2}} \dfrac{\ln(\tan x)}{\frac{1}{\cos x}} = \dfrac{-\infty}{\infty}$

$\dfrac{\frac{1+\tan^2 x}{\tan x}}{\frac{\sin x}{\cos^2 x}} = \dfrac{\frac{\sec^2 x}{\tan x}}{\sec x \cdot \tan x} = \dfrac{\sec x}{\tan^2 x} = \dfrac{\frac{1}{\cos x}}{\frac{\sin^2 x}{\cos^2 x}} = \dfrac{\cos x}{\sin^2 x}$

tru

$$\lim_{x\to\frac{\pi}{2}} \frac{\cos x}{\sin^2 x} = \frac{0}{1}$$

$$\ln y = 0 \qquad y = e^0 = 1$$

(48) $\lim_{x\to 0^+} \left(\frac{1}{x}\right)^{\sin x} = \infty^0$ bls

$$y = \left(\frac{1}{x}\right)^{\sin x} \Rightarrow \ln y = \sin x \cdot \ln\left(\frac{1}{x}\right)$$

$$\lim_{x\to 0^+} \left(\sin x \cdot \ln\left(\frac{1}{x}\right)\right) = 0 \cdot \infty \to \lim_{x\to 0^+} \frac{\ln\left(\frac{1}{x}\right)}{\frac{1}{\sin x}} = \frac{\infty}{\infty}$$

$$\sin^{-1} = -1 \cdot \sin^{-2} x \cdot \cos x$$

$$\frac{\frac{-1}{x^2}}{\frac{-\cos x}{\sin^2 x}} = \frac{\sin^2 x}{x \cdot \cos x} \Rightarrow \frac{0}{0} \to \frac{2\sin x \cdot \cos x}{\cos x - x \cdot \sin x} = \frac{0}{1} = 0$$

$$\ln y = 0 \qquad e^0 = 1 = y$$

(49) $\lim_{x\to 1} \left(\frac{x}{x-1} - \frac{x}{\ln x}\right) = ?$

$$\frac{x \cdot \ln x - x(x-1)}{(x-1)\ln x} = \frac{x\ln x - x^2 - x}{x\ln x - \ln x} \to x\ \frac{\cancel{1}\ln x - 2x \cancel{-1}}{1 + \ln x - \frac{1}{x}} = \frac{-2}{0}$$

$$\frac{-1}{2} = \frac{\frac{1}{x} - 2}{\frac{1}{x} + \frac{1}{x}}$$

20a) $\lim_{x\to\infty}\left(e^{\frac{1}{x}}+\frac{1}{x}\right)^{x} = 1^{\infty}$

$y=\left(e^{\frac{1}{x}}+\frac{1}{x}\right)^{x}$ $\quad \ln y = x\cdot \ln\left(e^{\frac{1}{x}}+\frac{1}{x}\right)$

$\lim_{x\to\infty} x\cdot \ln\left(e^{\frac{1}{x}}+\frac{1}{x}\right) = \infty\cdot 0 \to \lim_{x\to\infty}\frac{\ln\left(e^{\frac{1}{x}}+\frac{1}{x}\right)}{\frac{1}{x}} = \frac{0}{0}$

$t=\frac{1}{x}$, $x\to\infty$ $t\to 0$

$\lim_{t\to 0}\frac{\ln(e^t+t)}{t}$

$\frac{\frac{e^t+1}{e^t+t}}{1} \Rightarrow \lim_{t\to 0}\left(\frac{e^t+1}{e^t+t}\right) = \frac{2}{1}$

$\ln y = 2 \quad e^2 = y$

20b) $\lim_{x\to 0^+}\left(\cot x - \frac{1}{x}\right) = \infty-\infty$

$\frac{x\cdot\cot x - 1}{x} = \frac{x\cos x - \sin x}{x\cdot\sin x} = \frac{0}{0} \to \frac{\cos x - x\sin x - \cos x}{\sin x + x\cos x}$

$\lim_{x\to 0^+}\frac{-x\sin x}{\sin x + x\cos x} = \frac{0}{0} \to \frac{-\sin x - x\cos x}{\cos x + \cos x - x\sin x} = \frac{0}{2} = 0$

21) $\lim_{x\to 0}(1+x+2x^2+3x^3+\dots)^{1/2x} = 1^{\infty}$

$y = (1+x+2x^2+3x^3\dots)^{1/2x}$ $\quad \ln y = \frac{\ln(1+x+2x^2\dots)}{2x}$

$\lim_{x\to 0}\frac{\ln(1+x+2x^2+\dots)}{2x} = \frac{0}{0} \to \frac{\left(\frac{1+4x+9x^2+\dots}{1+x+2x^2\dots}\right)}{2} = \frac{1}{2}$

$\ln y = \frac{1}{2}$ $\quad y = e^{\frac{1}{2}}$

22) $\lim_{x\to\infty}\frac{2x^2+5}{\ln(\ln(x^2))} = \frac{\infty}{\infty}$

$\lim_{x\to\infty}\frac{4x}{\frac{2}{x\cdot\ln(x^2)}} = \lim_{x\to\infty} 2x^2\ln(x^2) = \infty\cdot\infty = \infty$

23) $\lim_{x\to 0}\frac{\ln(x)}{\cot(x)} = \frac{-\infty}{\infty}$

$(\ln(x))' = \frac{1}{x}$

$(\cot(x)) = -\operatorname{cosec}^2 x$

$\lim_{x\to 0}\frac{\frac{1}{x}}{\frac{-1}{\sin^2}} = \lim_{x\to 0}\left(\frac{\sin^2 x}{-x}\right)$

$\operatorname{cosec} = \frac{1}{\sin}$

$\lim_{x\to 0}\frac{\sin x}{x}\cdot\lim_{x\to 0}\frac{\sin x}{-1}$

$1 \cdot 0 = 0$

24a) $\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{\sin x}\right) = \infty-\infty$ bls.

$\lim_{x\to 0}\frac{\sin x - x}{x\cdot\sin x} = \frac{0}{0} \to \lim_{x\to 0}\frac{\cos x - 1}{\sin x + x\cos x} = \frac{0}{0} \to \frac{-\sin x}{\cos x + \cos x - x\sin x}$

$= \frac{0}{2} = 0$

24b) $\lim_{x\to 0} x^{\sin x} = 0^0$ bls.

$y = x^{\sin x} \xrightarrow{\ln} \ln y = \sin x \cdot \ln x$

$\lim_{x\to 0} \sin x \cdot \ln x = 0\cdot\infty \to \lim_{x\to 0}\frac{\ln x}{\frac{1}{\sin x}} = \frac{\infty}{\infty} \to \frac{\frac{1}{x}}{\frac{-\cos x}{\sin^2 x}} = \frac{\sin^2 x}{-x\cdot\cos x}$

$\left(\frac{1}{\sin x} = \operatorname{cosec} x\right)'$

$-\operatorname{cosec} x\cdot\cot x$

$\frac{-\cos x}{\sin^2 x}$

$-\frac{0}{0}$

$\frac{2\sin x\cdot\cos x}{-\cos x + x\sin x}$

$\ln y = \frac{0}{-1} = 0$

$\ln y = 0$

$y = e^0 = 1$

24c) $\lim_{x\to 0^+} x^2\cdot\ln x = 0\cdot(-\infty)$

$\lim_{x\to 0^+}\frac{\ln x}{\frac{1}{x^2}} \to \frac{\frac{1}{x}}{\frac{-2}{x^3}} = \frac{x^2}{-2} = 0$

$(x^{-2})' = -$

25) $\sqrt{18}-\sqrt{17}$, $\sqrt{17}-\sqrt{16}$ hang daha büyük?

$$\sqrt{18}-\sqrt{17} = \frac{(\sqrt{18}-\sqrt{17})\cdot(\sqrt{18}+\sqrt{17})}{(\sqrt{18}+\sqrt{17})} = \frac{1}{\sqrt{18}+\sqrt{17}}$$

$$\sqrt{17}-\sqrt{16} = \frac{(\sqrt{17}-\sqrt{16})(\sqrt{17}+\sqrt{16})}{(\sqrt{17}+\sqrt{16})} = \frac{1}{\sqrt{17}+\sqrt{16}}$$

$(\sqrt{18}+\sqrt{17}) > (\sqrt{17}+\sqrt{16})$ old. $(\sqrt{17}-\sqrt{16}) > (\sqrt{18}-\sqrt{17})$

26) $f(x) = \frac{x^3}{3} - x^2 - 3x + 18$ hem artan hem azalan

$f'(x) = x^2 - 2x - 3$

$x \quad -3$

$x \quad 1$

$x_1 = 3, \; x_2 = -1$

$f''(x) = 2x - 2 \qquad x_1 = 1$

$f(x)$	$-\infty$	-1	1	∞
$f'(x)$	$+$	$=$	$=$	$+$
$f''(x)$	$-$	$=$	$+$	$+$
	$\cap$	$\cap$	$\cup$	$\cup$

$(3, \infty)$

27) $y = x^3 + mx^2 + nx - 4$ $\quad f''(1) = 0,\ f'(1) = 2,\ n?$

$f'(x) = 3x^2 + 2mx + n$

$f'(1) = 3 + 2m + n = 2 \quad 2m + n = -1$

$n = 5$ /1

$f''(x) = 6x + 2m$

$f''(1) = 6 + 2m = 0 \quad m = -3$

28) $y = \frac{x}{x^2+1}$

$y' = \frac{1 - x^2}{(1+x^2)^2} = 0 \to x^2 = 1$

$x_1 = 1 \quad x_2 = -1$

y		-1		1	
y'	$-$	0	$+$	0	$-$
	↘	min	↗	max	↘

$y(-1) = 0$ $\quad (-1,0)$ ekstremum

$y(1) = 0$ $\quad (1,0)$ ekstremum

$(-\infty, -1] \cup [1, \infty)$ azalan

$[-1, 1]$ artan

$$y'' = \frac{-2x(1+x^2)^2 - (1-x^2)(4x)(1+x^2)}{(1+x^2)^4}$$

$$= \frac{(2x(1+x^2))(-(1+x^2) - (1-x^2)(2))}{(1+x^2)^4} = \frac{2x(x^2-3)}{(1+x^2)^3}$$

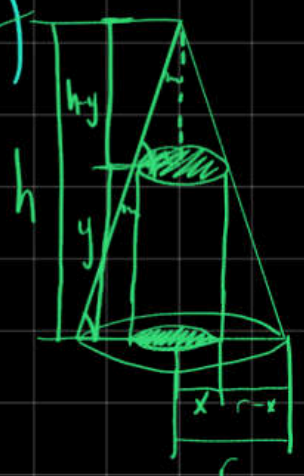
$2x = 0 \quad \cup \quad (x^2 - 3) = 0$

$x_1 = \sqrt{3} \quad x_2 = -\sqrt{3}$

Bükum $= -\sqrt{3}, 0, \sqrt{3}$

/1

29)



$V(x) = \pi x^2 \cdot y$

$\frac{h-y}{h} = \frac{x}{r}$ $\quad h-y = \frac{hx}{r}$ $\quad y = h - \frac{hx}{r}$

$y = h\left(1 - \frac{x}{r}\right)$

$V(x) = \pi x^2 \cdot h\left(1 - \frac{x}{r}\right) = \pi \cdot h \cdot \left(x^2 - \frac{x^3}{r}\right)$

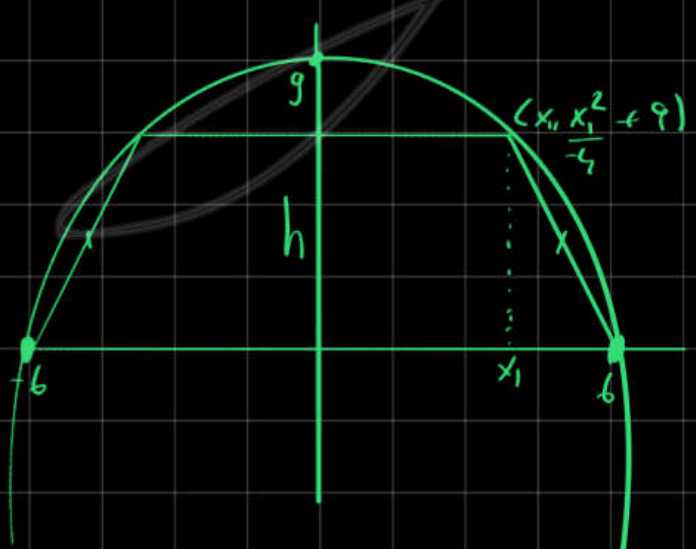
$V'(x) = \pi \cdot h \cdot \left(2x - \frac{3x^2}{r}\right) = 0$

$2x = \frac{3x^2}{r}$

$2 = \frac{3x}{r}$ $\quad x = \frac{2r}{3}$

$V_{max}\left(\frac{2r}{3}\right) = \pi \cdot \left(\frac{2r}{3}\right)^2 \cdot h \cdot \left(1 - \frac{\frac{2r}{3}}{r}\right) = \pi \cdot \frac{4r^2}{9} \cdot \frac{h}{3} = \frac{4r^2 h \pi}{27}$

30)



$x^2 + 4y - 36 = 0$

$-\frac{x^2}{4} + 9 = y$

$x_1 = 6 \quad x_2 = -6$

$A(x) = \frac{(12 + 2x_1)}{2} \cdot \left(\frac{x_1^2}{-4} + 9\right) = (6 + x)\left(\frac{x^2 - 36}{-4}\right)$

$= \frac{6x^2 - 216 + x^3 - 36x}{-4}$

$= (x^3 + 6x^2 - 36x - 216)/4$

$A'(x) = \frac{1}{4}(3x^2 + 12x - 36) \Rightarrow \frac{3}{4}x^2 + 3x - 9 = 0$

$\frac{x^2}{4} + x - 3 = 0$

$x^2 + 4x - 12 = 0$

$x \quad 6 \quad -6 = x_1$

$x \quad -2 \quad 2 = x_2$

$A(2) = 8\left(\frac{4 - 36}{-4}\right) = 8 \cdot 8 = 64$ m²

31)



$x \cdot y = 64$

$y = \frac{64}{x}$

$C(x) = 2x + 2y = 2x + \frac{128}{x}$

$C'(x) = 2 - \frac{128}{x^2} = 0$

$x^2 = 64$

$x = \pm 8$

$x = 8 \quad y = 8$

$C(8) = 16 + 16 = 32$ m

32) $y = x \cdot e^x$ nın eğimli teğet?

$y' = e^x + x \cdot e^x$

$y'' = e^x + e^x + x \cdot e^x = e^x(2 + x) \rightarrow e^x = 0 \quad \cup \quad x = -2$

$\emptyset$

$y'(-2) = \frac{-1}{e^2}, \quad y(-2) = \frac{-2}{e^2}$

$$y + \frac{2}{e^2} = \frac{-1}{e^2}(x+2)$$

33) $y = x^3 - 4x^2$

$y' = 3x^2 - 8x$

$y'' = 6x - 8 \quad x_1 = \frac{8}{6} = \frac{4}{3}$

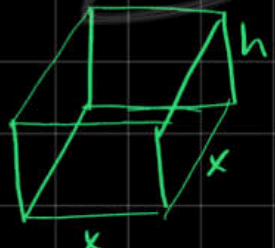
$y'\left(\frac{4}{3}\right) = \frac{48}{9} - \frac{96}{9} = -\frac{48}{9} = -\frac{16}{3}$

$y\left(\frac{4}{3}\right) = \left(\frac{4}{3}\right)^3 - 4\cdot\left(\frac{4}{3}\right)^2 = \left(\frac{16}{9}\right)\left(\frac{-8}{3}\right) = -\frac{128}{27}$

$m = \frac{-16}{3}, \quad \left(\frac{4}{3}, \frac{-128}{27}\right)$

$y + \frac{128}{27} = \left(\frac{-16}{3}\right)\left(x - \frac{4}{3}\right)$

34)



$x^2 \cdot h = 32 \quad h = \frac{32}{x^2}$

$x^2 + 4xh = x^2 + \frac{128}{x} = f(x)$

$f'(x) = \frac{2x^3 - 128}{x^2}$

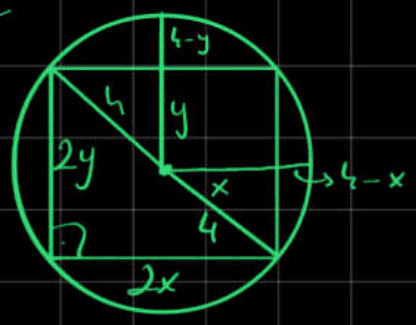
$x_1 = x_2 = 0 \quad x^3 = 64$

$x = 4$

x		0		4	
$f'(x)$	$-$	$\nexists$	$-$	0	$+$
$f(x)$	$\searrow$	$\nexists$	$\searrow$	min	$\nearrow$

$f(4) = -16 + \frac{128}{4} = 48$

35)



$4y^2 + 4x^2 = 64$

$y^2 + x^2 = 16$

$y^2 = 16 - x^2$

$y = \sqrt{16 - x^2}$



$a^2 + b^2 = 64$

1

$A(x) = 4xy = 4x\sqrt{16 - x^2}$

$A'(x) = 4\sqrt{16 - x^2} + 4x \cdot \frac{-2x}{2\sqrt{16 - x^2}} = 4\sqrt{16 - x^2} \; \frac{-4x^2}{\sqrt{16 - x^2}}$

$= \frac{4(16 - x^2) - 4x^2}{\sqrt{16 - x^2}} = \frac{64 - 8x^2}{\sqrt{16 - x^2}}$

$64 - 8x^2 = 0$

$8 - x^2 = 0$

$x^2 = 8$

$x_1 = 2\sqrt{2}$

$x_2 = -2\sqrt{2}$

x		$-2\sqrt{2}$		$2\sqrt{2}$	
$f'(x)$	−	0	+	0	−
	↘		↗		↘

$A(2\sqrt{2}) = 4 \cdot 2\sqrt{2} \cdot 2\sqrt{2} = 4 \cdot 2 \cdot 2 \cdot 2 = 32$

36) $f(x) = x^3 + ax^2 + bx$

$f'(x) = 3x^2 + 2ax + b$

$f'(-1) = 3 - 2a + b = 0 \quad b - 2a = -3$

$f'(3) = 27 + 6a + b = 0 \quad 6$

$-24 - 8a = 0$

$a = -3 \quad b = -9$

$f'(x) = 3x^2 + 2ax + b$

$f'(4) = 48 + 8a + b = 0$

$f''(x) = 6x + 2a$

$f''(1) = 6 + 2a = 0$

$a = -3 \quad b = -24$

(37a) Gonki 10 s ucar 10 s duser toplam 20 saniye ucus suresidir.

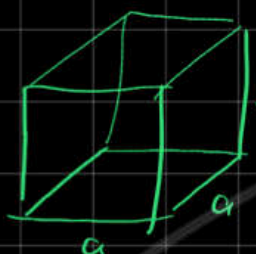
(37b) $h(t) = -16t^2 + 320t$

$h'(t) = -32t + 320$

$t_1 = 10$

$h(10) = -1600 + 3200 = 1600$

(38)



$a^2 \cdot b = 32$

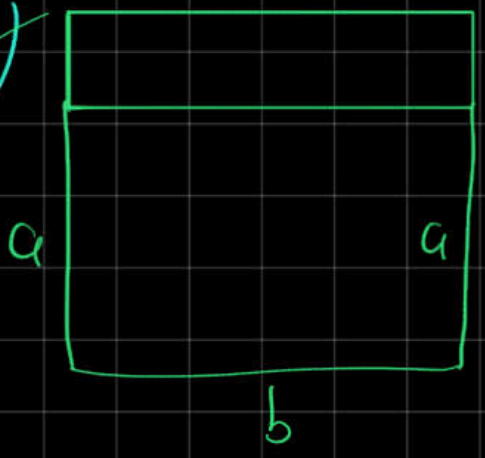
$b = \frac{32}{a^2}$

$A = a^2 + 4ab = a^2 + \frac{128}{a} \rightarrow 2a - \frac{128}{a^2} = \frac{2a^3 - 128}{a^2}$

$a^3 = 64$

$a = 4 \quad b = 2$

39)



$b = 800 - 2a$

$2a + b = 800$

$A(a) = a \cdot b = a(800 - 2a)$

$800a - 2a^2$

$A'(a) = 800 - 4a$

$a = 200$

$b = 400$ ↑

40)



$a \cdot b = 216$

$b = \frac{216}{a}$

$C(a) = 3a + \frac{432}{a}$

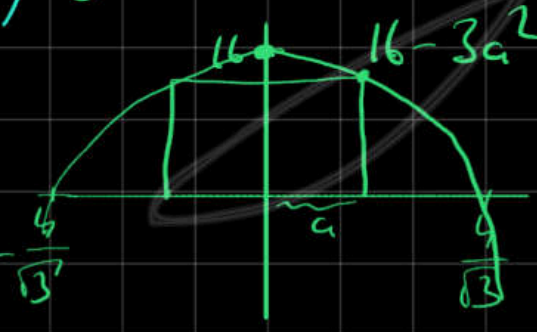
$\frac{3a^2 - 432}{a^2}$

$a^2 = 144$

$a = \pm 12$

$a = 12, b = 18$

41) $y = -3x^2 + 16$



$3x^2 = 16$

$x^2 = \frac{16}{3}$

$x = \pm \frac{4}{\sqrt{3}}$ $y = 16$

$A(a) = 2a(16 - 3a^2) = 32a - 6a^3$

$\frac{8}{3}\left(16 \cdot \frac{4^2}{9}\right) = \frac{32 \cdot 8}{9}$

$= \frac{256}{9}$ ↑

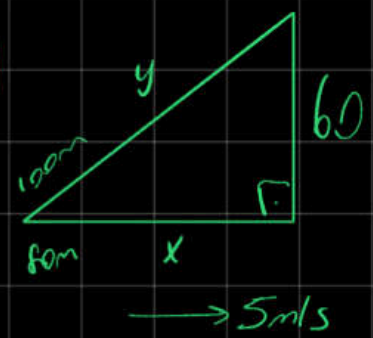
$32 - 18a^2 = 0$

$32 = 18a^2$

$16 = 9a^2$

$\frac{4}{3} = a$

42)



$x^2 + 60^2 = y^2$

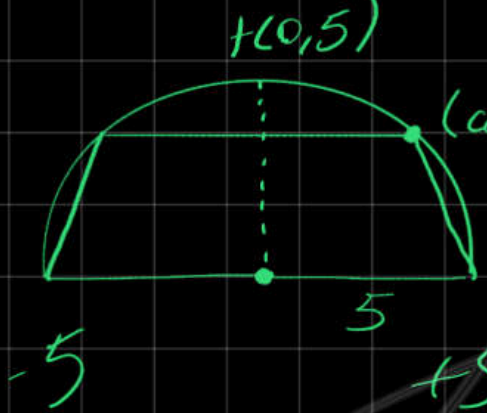
$\frac{dx}{dt} = 5 \quad \frac{dy}{dt} = ?$

$2x \cdot \frac{dx}{dt} = 2s \cdot \frac{ds}{dt}$

$2 \cdot 80 \cdot (5) = 2 \cdot (100) \cdot \frac{ds}{dt}$

$\frac{ds}{dt} = 4 \text{ m/s}$

43)



parabol $= a(x-r)^2 + k$

$= a(x-x_1)^2$

$= a(x-x_1)(x-x_2)$

$x^2 + y^2 = 25$

$(5+a)(25-a^2)$

$= 125 - 5a^2 + 25a - a^3$

$-3a^2 - 10a + 25 = 0$

$a^2 + y^2 = 25$

$y = \sqrt{25-a^2}$

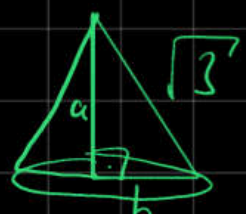
$\Delta = 100 - 4(-3 \cdot 25)$

$= 400$

$\frac{-b \pm \sqrt{\Delta}}{2a} \Rightarrow \frac{10+20}{-6} = -5 \quad \vee \quad \frac{10-20}{-6} = \frac{5}{2} = a$

$$25 - \frac{25}{5} = \frac{25 \cdot 3}{5} = \frac{75}{4}$$ 1

44)



$a^2 + b^2 = 3$

$b^2 = 3 - a^2$

$b = \sqrt{3 - a^2}$

$$\frac{\pi \cdot b^2 \cdot a}{3} = H(a)$$

$$\frac{\pi (3 - a^2) a}{3}$$

$$= \frac{\pi (3a - a^3)}{3}$$ 1

$$H(1) = \frac{\pi 2 \cdot 1}{3} = \frac{2\pi}{3}$$ 1

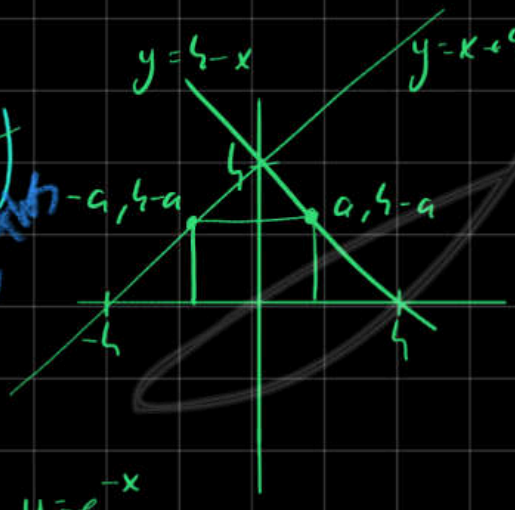
$$\frac{\pi}{3} \cdot (3 - 3a^2)$$

$$= \pi (1 - a^2) = 0$$

$a^2 = 1$

$a = \pm 1 \quad a = 1$ 1

45)



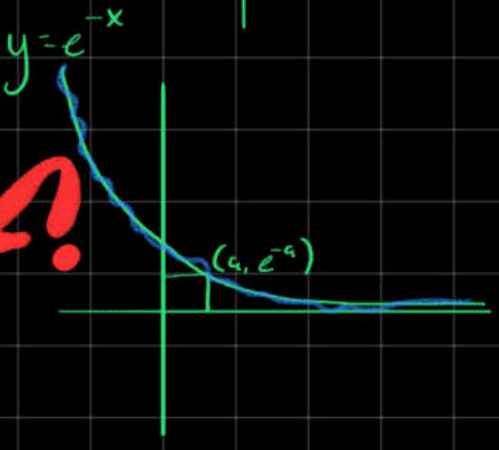
$2a(4 - a) = 8a - 2a^2$ 1

$8 - 4a$

$a = 2$

$16 - 8 = 8$ 1

46)

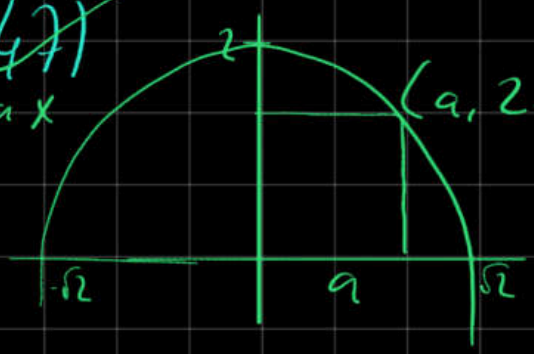


$A(a) = a \cdot e^{-a}$

$A'(a) = e^{-a} - a \cdot e^{-a} = \frac{1}{e^a}(1 - a) = 0$

$a = 1$

$A(1) = e^{-1} = \frac{1}{e}$ m

47) 

$y = -x^2 + 2 \quad y' = -2x$

$A(a) = a(2-a^2) = 2a - a^3$

$2 - 3a^2$

$2 = 3a^2$

$\sqrt{\frac{2}{3}} = a$

$y = 2 - \frac{2}{3} = \frac{4}{3} \quad m = -2\sqrt{\frac{2}{3}}$

$y - \frac{4}{3} = -2\sqrt{\frac{2}{3}}\left(x - \sqrt{\frac{2}{3}}\right)$

48) $Q(t) = \frac{5t}{t^2+400}$ max?

not bille hoş

$Q' = \frac{5(t^2+400) - 5t(2t)}{(t^2+400)^2} = \frac{2000 - 5t^2}{(t^2+400)^2}$

$2000 = 5t^2$

$400 = t^2 \quad (t = 20)$

a) $\lim_{x \to \infty} \frac{5t}{t^2+400} = 0$ a yaklaşır

49) $y = \frac{ax+4}{bx-6}$ yatay asy $y=3$, düşey asy. $x=-3$ ise a.b?

düşey → tanımsızlık $\Big\}$ $bx - 6 \to -3b - 6 = 0$

$b = -2$

yatay → limit { $\lim_{x\to\infty} \frac{ax+4}{-2x-6} = -\frac{a}{2} = 3$

$a = -6$

$a \cdot b = 12$

50) $f(x) = 3x^4 - 4x^3 + 5$ grafiği?

$x=0, y=5$
$x=1, y=3-4+5=4$
$x=-1, y=3+4+5=12$
$x=-2, y=48+32+5=+$
$x=2, y=48-32+5=+$
$x=3, y=3^4-4.3^3+5 = 243-108+5=+$
$x=3, y=+$
$x=4, y=3.4^4-4^4+5=+$
$x=4, y=+$
$x=5, y=3.5^4-4.5^3+5=+$ (500)

① $f\in = R$

② $x=0, y=5$
$y=0, x= \rightarrow 3x^4-4x^3+5=0$ $\emptyset$

③ $f'(x) = 12x^3-12x^2 = 12x^2(x-1)=0$

$x_1=x_2=0 \quad x_3=1$

x		0		1	
f'	−	||	−	0	+
	↘		↘	min	↗

$\min f(1) = 3-4+5=4$

④ Konkavlik genelemiyor

⑤ tanımsız yapan yok → düşey asp yok

$\lim_{x\to\infty}(3x^4-4x^3+5)=\infty$

$\lim_{x\to-\infty}(3x^4-4x^3+5)=-\infty$

} yatay asp yok

Polinom old. için asy. yok. ??

⑥

x		0		1	
f'	−	‖	−	0	+
	↘		↘		↗

⑦



5) $f(x) = \frac{2}{x-3}$

① $Dk = R - \{3\}$

② $x = 0 \rightarrow y = -\frac{2}{3}$

$y = 0 \rightarrow x = \emptyset$

③ $f(x) = \frac{2}{x-3} = 2.(x-3)^{-1}$ $f'(x) = \frac{2}{(x-3)^2} = 0$ $\emptyset$

max, min yok

④ Konkavlık gerekmez.

$f'(x) = 2.(x-3)^{-2}$ $f''(x) = \frac{-4}{(x-3)^3}$ $\emptyset$

⑤ $x=3$ tanımsız yapt için düşey asy.

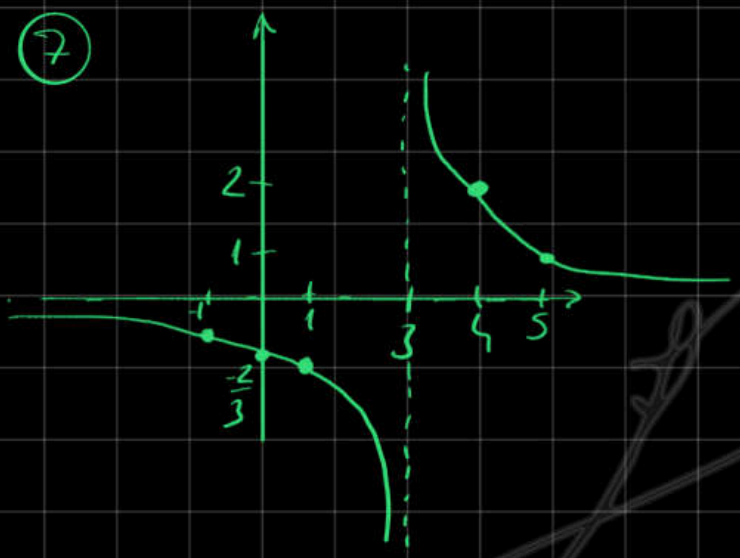
$\lim_{x \to \pm\infty} \frac{2}{x-3} = 0$ } $y=0$

yatay asy old. eğik/eğri asy yoktur

⑥

x		3	
f	↘		↘

⑦



52) $y = \frac{1}{x} - \frac{1}{x^2}$

① $tk = R - \{0\}$

② $f(x) = \frac{x-1}{x^2}$ $\quad x=0 \rightarrow \emptyset$ $\quad y=0 \rightarrow x=1$

③ $f'(x) = \frac{1 \cdot x^2 + (x-1)(-2x)}{x^4} = \frac{x(-x+2)}{x^4} = \frac{-x+2}{x^3}$

$-x+2=0$

$x=2 \qquad f(2)=\frac{1}{4}$

④ Karkacaiti gerekmiyor.

⑤ $f(x)=\frac{x-1}{x^2}$

$x=0$ düşey asp

$\lim_{x\to 0^+} \frac{x-1}{x^2} = \frac{x(1-\frac{1}{x})}{xx} = \infty$ ✓

$\lim_{x\to\infty} \frac{x-1}{x^2} = 0 \rightarrow$ yatay asp $=0$

$y=ax+b \qquad a=\lim_{x\to\infty} \frac{f(x)}{x} \qquad b=\lim_{x\to\infty} f(x)-ax$

$\lim_{x\to\infty} \frac{x-1}{x^3} = 0 \qquad \lim_{x\to\infty} \frac{x-1}{x^2} = 0 \qquad \boxed{y=0}$

$\lim_{x\to-\infty} \frac{x-1}{x^3} = 0$

$-2 \qquad -1$

$-3 \qquad -2$

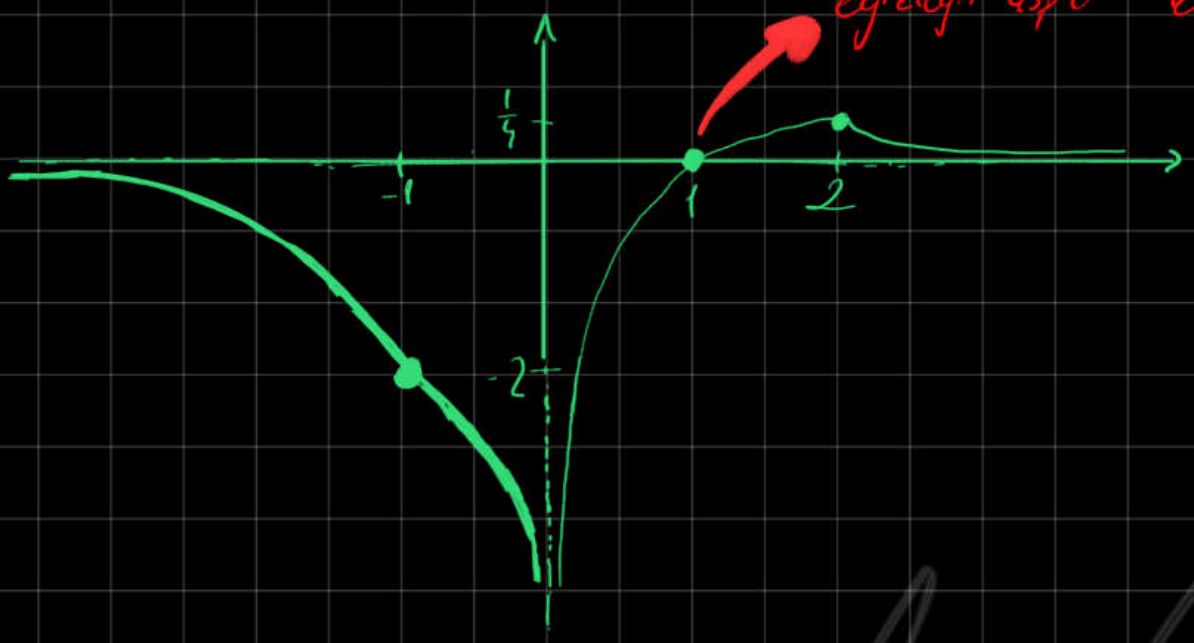
4

⑥

x		0		2	
$f'(x)$	$-$	$\exists$	$+$	$\exists$	$-$
$f(x)$	$\searrow$		$\nearrow$		$\searrow$

⑦

grafik düşey asimptot kesemez
yatay asimptot kesebilir
eğik/eğri asimptot kesebilir



53) $y = \frac{x}{x^2-1}$

① $Df = R - \{-1, 1\}$

② $x = 0 \rightarrow y = 0$

③ $\frac{1(x^2-1) - (2x)(x)}{(x^2-1)^2} = \frac{-(x^2+1)}{(x^2-1)^2} = 0$ ∅

④ Konkavlık gerekip gerekmediğini bilmiyorum

⑤ $x = -1$ ve $x = 1$ tanımsız yapayı

$\lim_{x \to (-1)^+} \frac{x}{x^2-1} = \frac{-0,9}{-0,1} = \infty$ $\lim_{x \to 1^+} \frac{x}{x^2-1} = \frac{1,1}{0,1} = \infty$

$\lim_{x \to (-1)^-} \frac{x}{x^2-1} = \frac{-1,1}{0,1} = -\infty$ $\lim_{x \to 1^-} \frac{x}{x^2-1} = \frac{0,9}{-0,1} = -\infty$

$\lim_{\substack{x\to\infty\\ -\infty}} \frac{x}{x^2-1} = \frac{\infty}{\infty} \to \frac{1}{2x} = \frac{1}{\infty} = 0$ $\boxed{y=0}$

⑥

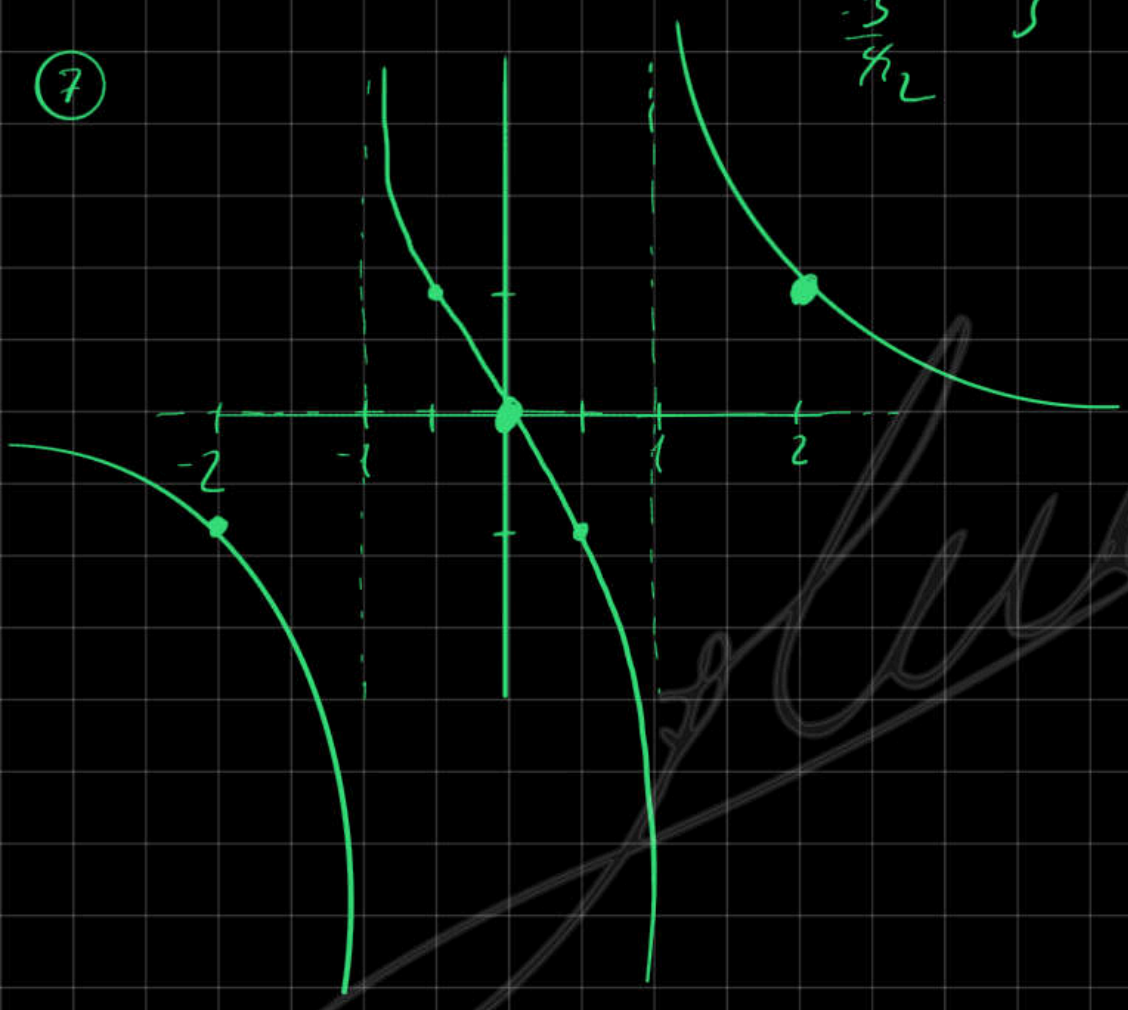
x				
f'(x)				
f(x)				

$f(2) = \frac{2}{3}$ $f(-2) = \frac{-2}{3}$

$f(\frac{1}{2}) = \frac{\frac{1}{2}}{-3} = \frac{-2}{3}$

x_2

⑦



54) $\frac{2x^2}{x^2-4} = f(x)$

① $H_f = R - \{-2, 2\}$

② $x=0 \to y=0$

$4x^3 - 4x^2 - 4x^3$

③ $f'(x) = \frac{4x(x^2-4) - (2x^2)(2x)}{(x^2-4)^2} = \frac{-4x^2}{(x^2-4)^2} = 0$ $x_{1,2} = 0$

④ Konkav ?

⑤ $f(x) = \frac{2x^2}{x^2-4}$

$x=2$, $x=-2$ tanımsız yapar

$\lim_{x \to 2^+} \frac{2x^2}{x^2-4} \xrightarrow{2^+} \frac{8}{0} \xrightarrow{} \frac{+}{+} = \infty$ $\quad \lim_{x \to 2^-} f(x) = -\infty$

$\lim_{x \to -2^+} f(x) \xrightarrow{-2} \frac{8}{0} \xrightarrow{-2^+} \frac{+}{-} = -\infty$ $\quad \lim_{x \to -2^-} f(x) = \infty$

$\lim_{x \to \infty} \frac{2x^2}{x^2-4} = \frac{x^2(2)}{x^2(1-\frac{4}{x^2})} = 2$ $\quad y=2$

$\lim_{x \to -\infty} \frac{2x^2}{x^2-4} = 2$

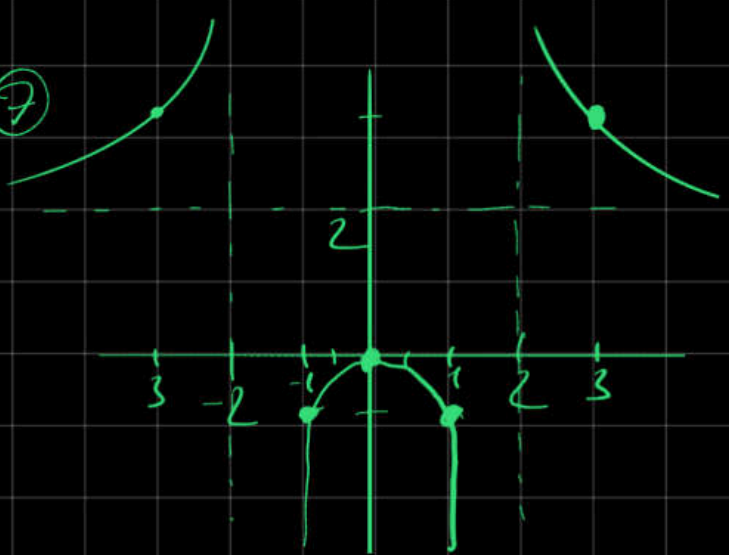
⑥

x		0		
f'(x)	−	‖	−	

$f(3) = \frac{18}{5}$ $\quad f(-3) = \frac{18}{5}$

$f(1) = \frac{2}{-3}$

⑦



55) $y = \frac{\sqrt{4-x^2}}{x}$

$4-x^2 \geq 0$

$-2 \quad 2$

$-$	$+$	$-$
-2	2	

① $tk = [-2, 2] - \{0\}$

② $x \neq 0$

$y = 0 \rightarrow x_1 = 2 \quad x_2 = -2$

③ $f'(x) = \frac{\frac{-x^3}{x\sqrt{4-x^2}} - (4-x^2)}{x^2} = \frac{\frac{-x^3+x^2-4}{x\sqrt{4-x^2}}}{x^2}$

$\frac{-x^3+x^2-4}{x\sqrt{4-x^2}} = 0 \quad \emptyset$

④ Konkav?

⑤ $f(x) = \frac{\sqrt{4-x^2}}{x}$

$x = 0$ düşey asp

$\lim_{x \to 0^+} \frac{\sqrt{4-x^2}}{x} = \frac{2}{0} \Rightarrow \frac{+}{+} = \infty$

$\lim_{x \to 0^-} f(x) \Rightarrow \frac{+}{-} = -\infty$

Sonsuza gidemez çünkü mutlu 2

$\lim_{x \to \infty} \frac{\sqrt{4-x^2}}{x} = \frac{\sqrt{x^2(\frac{4}{x^2}-1)}}{x} = \frac{|x|\sqrt{(\frac{4}{x^2}-1)}}{x} = 1$

$$\lim_{x\to-\infty}\frac{\sqrt{4-x^2}}{x}=\frac{|x|\sqrt{\left(\frac{4}{x^2}-1\right)}}{x}=-1$$

⑥

⑦



56) $y=e^{1/x}$

① $H_f = R-\{0\}$

② $x\neq 0$
$y\neq 0$

2

③ $(e^{1/x})' = \frac{-e^{1/x}}{x^2} = 0 \qquad -e^{1/x}=0 \qquad x$
$\varnothing$

④ $f'(x) = \frac{-e^{1/x}}{x^2} \qquad f''(x) = \frac{e^{1/x}\,x^2-(2x)(-e^{1/x})}{x^4}$

$= \frac{e^{1/x}+2x\cdot e^{1/x}}{x^4}$

$= e^{1/x}(1+2x) = 0$

$x = -\frac{1}{2}$

→ Türev değişiyorsa dönüm noktası

$f''(-\frac{1}{2}) = 0$

⑤ $e^{1/x}$

$x = 0$ tanımsız

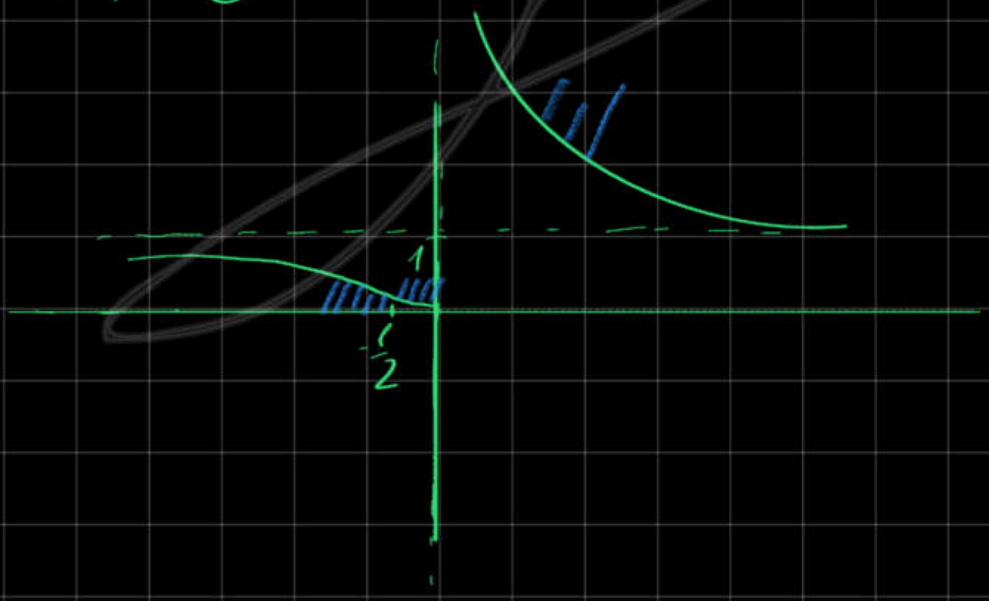
$\lim_{x \to 0^+} e^{1/x} = e^{\infty} \to \infty$ $\lim_{x \to 0^-} e^{1/x} = e^{-\infty} = 0$

$\lim_{x \to \infty} e^{1/x} = e^0 = 1$ $\lim_{x \to -\infty} e^{1/x} = e^0 = 1$ $y = 1$

⑥

x		$-\frac{1}{2}$	
f''	−	0	+
	∩		∪

⑦



57) $y = e^{1+\frac{1}{x}}$

① $\mathbb{R} = R - \{0\}$

② $x \neq 0$
$y \neq 0$

③ $f(x) = e^{1+\frac{1}{x}}$ $\quad f'(x) = \frac{-e^{1+\frac{1}{x}}}{x^2} = 0$
$\emptyset$

④ $f''(x) = \frac{e^{1+\frac{1}{x}} \cdot x^2 + (2x)(+e^{1+\frac{1}{x}})}{x^4} = \frac{e^{1+\frac{1}{x}}(1+2x)}{x^4} = 0$

$x = -\frac{1}{2}$

⑤ $y = e^{1+\frac{1}{x}}$

$x = 0$ İmsiz yapar

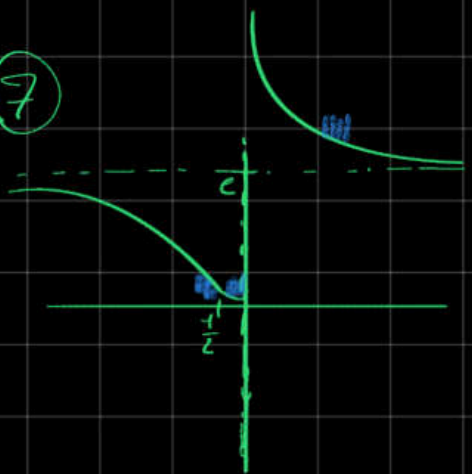
$\lim_{x \to 0^+} e^{1+\frac{1}{x}} = e^{\infty} = \infty$ $\quad \lim_{x \to 0^-} e^{1+\frac{1}{x}} = e^{-\infty} = 0$

$\lim_{x \to \infty} e^{1+\frac{1}{x}} = e$ $\quad \lim_{x \to -\infty} e^{1+\frac{1}{x}} = e$ $\quad y = e$

⑥

x		$-\frac{1}{2}$	
f''	$-$	$\|$	$+$
	∩		∪
	konkav		konveks

⑦



58) $f(x) = \left(\frac{x^2+2x}{x^2-1}\right)$

① $Dk = R - \{-1, 1\}$

② $x = 0 \to \frac{0}{-1} \Rightarrow y = 0$

③ $f(x) = \left(\frac{x^2+2x}{x^2-1}\right)$ $f'(x) = \frac{(2x+2)(x^2-1) - (2x)(x^2+2x)}{(x^2-1)^2}$

$= 2x^3 - 2x + 2x^2 - 2 - 2x^3 - 4x^2 \Rightarrow \frac{-2(x^2+x+1)}{(x^2-1)^2}$

$x^2 + x + 1 = 0$

$\Delta = 1 - 4 = -3$ $\emptyset$

④ $f''(x) =$

⑤ $x = 1$, $x = -1$ disco asp.

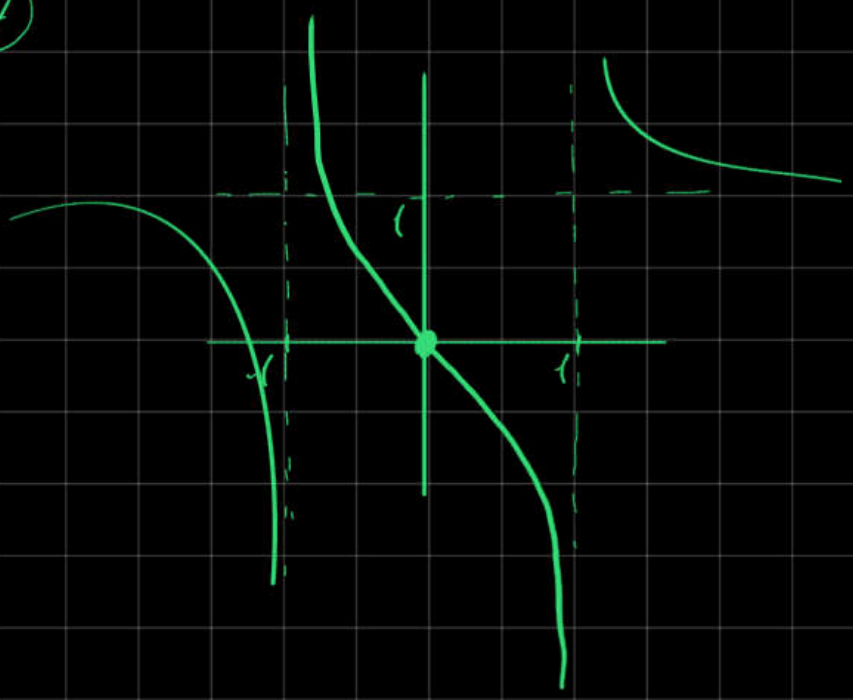
$\lim_{x \to 1^+} \frac{x^2+2x}{x^2-1} = \frac{3}{0} \to \infty$ $\lim_{x \to 1^-} \frac{x^2+2x}{x^2-1} \to -\infty$

$\lim_{x \to -1^+} \frac{x^2+2x}{x^2-1} = \frac{-1}{0} \to \lim_{x \to -1^+} \frac{(-0,9)^2 + 2(-0,9)}{(0,9)^2 - 1} = \frac{-}{-} = \infty$

$\lim_{x \to -1^-} f(x) = -\infty$

$\lim_{x \to \infty} \frac{x^2(1+\frac{2}{x})}{x^2(1-\frac{1}{x^2})} = 1$ $\lim_{x \to -\infty} = 1$ $y = 1$

(7)



59) $f(x) = \ln\left(\frac{x-3}{x+3}\right)$ $\quad -\ 7\ +$

① $\frac{x-3}{x+3} > 0$ $\quad$ $\begin{array}{c|ccc} x & -3 & 3 \\ \hline + & \| & - \\ + & \| & + \end{array}$ $\quad D_f = R - [-3,3]$

② $x \neq 0$
$y \neq 0 \rightarrow \ln(1)$ $\quad x-3 = x+3$

③ $\ln\left(\frac{x-3}{x+3}\right) \rightarrow \frac{1(x+3) + (1)(-x+3)}{(x+3)^2}$ $\quad \frac{x-3}{x+3}$ $\quad = \frac{6}{(x+3)(x-3)}$

$6(x^2-9)^{-1}$ $\quad \frac{1}{\frac{6}{x^2-9}} \neq 0$

④ $f''(x) = \frac{-12x}{(x^2-9)^2}$ $\quad f''(0) = 0$ yaklıkta değil

⑤ tanımsız yapan $x = -3$ ve $x = 3$

$$\lim_{x \to -3} \ln\left(\frac{x-3}{x+3}\right) = \frac{-6}{0} \to \ln(-\infty) = 0$$

Tavim orediğine dahil değil.

$$\lim_{x \to -3^-} \ln\left(\frac{x-3}{x+3}\right) \to +\ln(\infty) = \infty$$

$$\lim_{x \to 3^+} \ln\left(\frac{x-3}{x+3}\right) = \ln(0) \to \ln(0^+) = -\infty$$

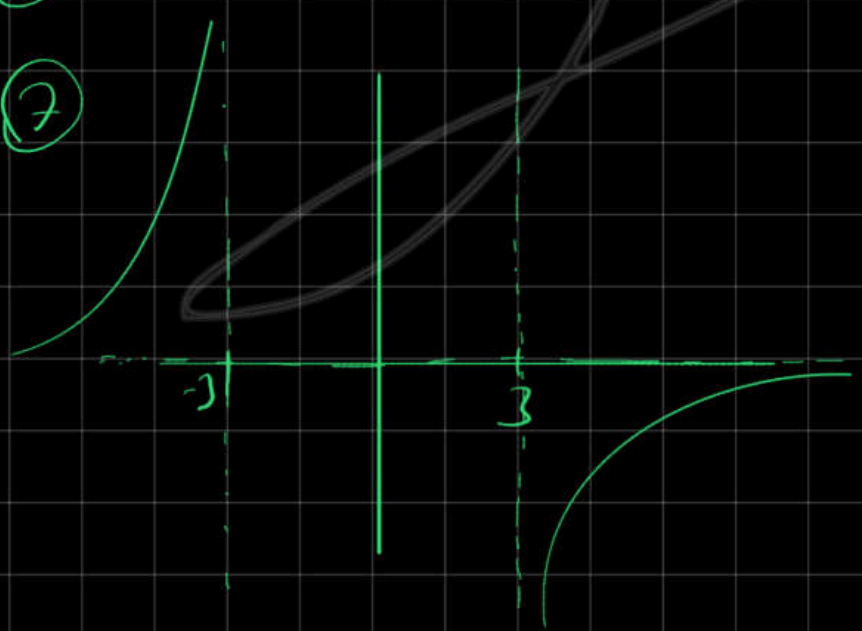
$\frac{0,1}{6,1} = 0^+$

$$\lim_{x \to 3^-} \ln\left(\frac{x-3}{x+3}\right) = \ln(-\infty) = 0$$

$$\lim_{x \to \infty} \ln\left(\underbrace{\frac{x-3}{x+3}}_{1}\right) = \ln 1 = 0 \quad (y = 0)$$

⑥

⑦



(60) $f(x) = \frac{\ln x}{x^2}$

① $x > 0 \quad x_{1,2} = 0 \quad x_3 = 1$

$tk : (0, \infty)$

x	0	1
	∄	− 0 +

② $x \neq 0$

$y = 0 \rightarrow \ln x = 0 \leftarrow x = 1$

③ $f(x) = \frac{\ln x}{x^2} \quad f'(x) = \frac{x - 2x \ln x}{x^4} = \frac{x(1 - 2\ln x)}{x^4}$

$1 - 2\ln x = 0$

$1 = 2\ln x$

$\frac{1}{2} = \log_e x \quad e^{1/2} = x = \sqrt{e}$

④

⑤ $x = 0$ tanımsız yapıyor ama tanım aralığına dahil değil

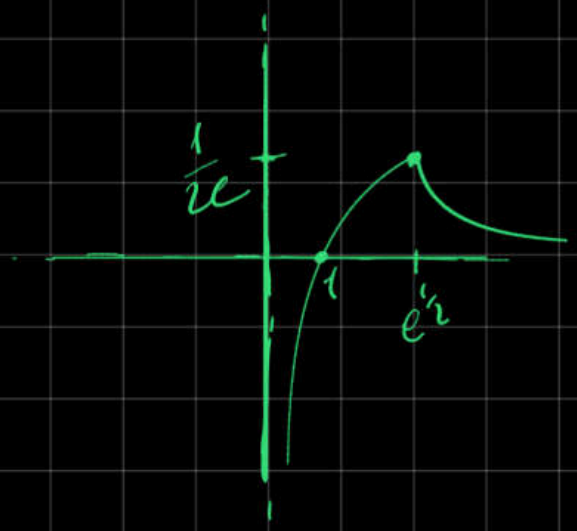
$\lim_{x \to 0^+} \frac{\ln x}{x^2} = \frac{1}{0} \rightarrow \frac{\ln(0^+)}{+} = -\infty$

$\lim_{x \to \infty} \frac{\ln x}{x^2} = 0 \quad (y = 0)$

⑥

x	0	$e^{1/2}$
f'	∄ +	−

(7)



60c) $f(x) = \frac{e^x}{e^x+1}$

① $e^x \neq -1 \quad \forall x = R$

② $x = 0, \; y = \frac{1}{2}$

$y \neq 0$

③ $f(x) = \frac{e^x}{e^x+1} \quad f'(x) = \frac{e^x(e^x+1) - (e^x)(e^x)}{(e^x+1)^2}$

$e^x \cdot e^x + e^x - e^x \cdot e^x$

$= \frac{e^x}{(e^x+1)^2} \neq 0$

④ $\frac{e^x}{(e^x+1)^2} \rightarrow \frac{e^x(e^x+1)^2 - 2(e^x+1)(e^x)(e^x)}{(e^x+1)^4}$

$= \frac{e^x(e^x+1)[(e^x+1) - 2e^x]}{(e^x+1)^4}$

$$= \frac{e^x(1-e^x)}{(e^x+1)^3} = 0 \qquad 1 = e^x \qquad x = 0$$

x		0	
f''	$+$	0	$-$
	$\cup$		$\cap$

(5) $\frac{e^x}{e^x+1}$

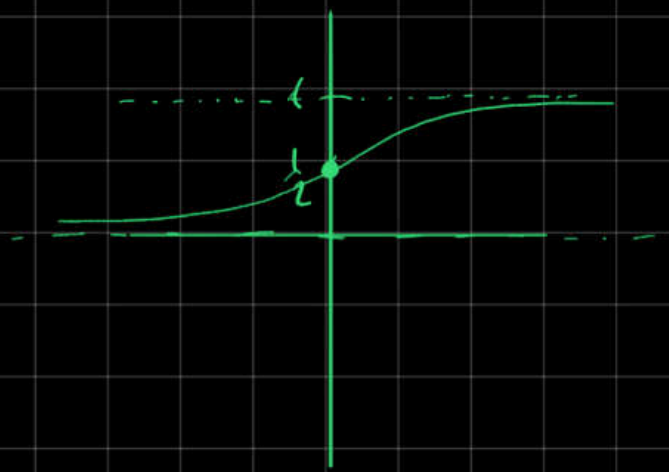
Tanımlı yapan yok

$$\lim_{x\to\infty} \frac{e^x}{e^x+1} \to \frac{e^x}{e^x} = 1 \qquad y = 1$$

$$\lim_{x\to-\infty} \frac{e^x}{e^x+1} = \frac{0}{1} = 0 \qquad y = 0$$

(6)

(7)



6(e) $f(x) = \frac{\ln x}{\ln(x-1)}$

① $x=2 \quad \ln(1)=0$ $\qquad$ $\underline{\text{Df} = (1,\infty) - \{2\}}$

$x>0 \qquad x-1>0$

$x>1$

② $y=0$, ~~$x=1$~~

④ tanımsız yapan $(x=2)$

$$\lim_{x\to 2^+} \frac{\ln x}{\ln(x-1)} = \frac{\ln(2)}{0} \to \frac{+}{+} = +\infty$$

$$\lim_{x\to 2^-} \frac{\ln x}{\ln(x-1)} \to \frac{\ln(2^-)}{\ln(1^-)} = \frac{+}{-} = -\infty \quad ?$$

$$\lim_{x\to\infty} \frac{\ln x}{\ln(x-1)} = \frac{\infty}{\infty} \xrightarrow{\text{l'hospital}} \frac{x-1}{x} = 1 \qquad (y=1)$$

$$\lim_{x\to-\infty} \frac{\ln x}{\ln(x-1)} = \frac{\ln(-\infty)}{\ln(-\infty)} = \frac{0}{0} \to \frac{x-1}{x} = 1$$

③ $\frac{\ln(x)}{\ln(x-1)} \xrightarrow{'} \frac{\frac{1}{x}\cdot\ln(x-1) - \left(\frac{1}{x-1}\right)(\ln x)}{(\ln(x-1))^2}$

$$\frac{\left(\frac{\ln(x-1)}{x} - \frac{\ln x}{x-1}\right)}{(\ln(x-1))^2} = 0 \text{ TÇM}$$

(5) f''

(6)

(7)



62) $f'(x) = f^2(x) \cdot x^2$

$\frac{df}{dx} = f^2 x^2 \rightarrow \frac{df}{f^2} = x^2 dx \rightarrow f^{-2} df = x^2 dx$?

$= \int f^{-2} df = \int x^2 dx$

$\frac{-1}{f} = \frac{x^3}{3} + c$

$f(2)$

$\frac{-1}{3} = \frac{8}{3} + c \quad \} c = -3$

$\frac{-1}{f} = \frac{x^3}{3} - 3 = \frac{x^3 - 9}{3}$

$f(x) = \frac{-3}{x^3 - 9}$

$f(1) = \frac{-3}{-8} = \frac{3}{8}$ e

64) $\frac{dy}{dt} = e^t \sin(e^t - 2)$, $y(\ln 2) = 0$

$dy = e^t \sin(e^t - 2)\, dt$

$\int dy = \int e^t \cdot \sin(e^t - 2)\, dt$

$\begin{pmatrix} u = e^t - 2 \\ du = e^t \cdot dt \end{pmatrix} \rightarrow \int \sin u \, du = -\cos u + c = -\cos(e^t - 2) + c$

$y = -\cos(e^t - 2) + c$

$y(\ln 2) = 0 = -\cos(e^{\ln(2)} - 2) + c = -\cos 0 + c = -1 + c$

$c = 1$

$y = -\cos(e^t - 2) + 1$

65) $\int \frac{\sqrt{x^2 - 4}}{x^4} dx = ?$



$\sec\alpha = \frac{x}{2}$

$\cos\alpha = \frac{2}{x}$

$\tan\alpha = \frac{\sqrt{x^2-4}}{2}$

$\cot\alpha = \frac{2}{\sqrt{x^2-4}}$

$\left(\begin{array}{l} x = \frac{2}{\sin\alpha} \\ dx = 2 \cdot -\operatorname{cosec}\alpha \cdot \cot\alpha \cdot d\alpha \end{array} \right) \rightarrow -2 \int \frac{\sqrt{\frac{4}{\sin^2\alpha} - 4}}{\frac{16}{\sin^4\alpha}} \cdot \operatorname{cosec}\alpha \cdot \cot\alpha \cdot d\alpha$

$$-2\int \frac{\sqrt{\frac{4}{\sin^2 x}-4}}{16\csc^{4/3}\frac{x}{2}}\cdot \csc\alpha\cdot\cot\alpha\cdot d\alpha =$$

$$\left(x=\frac{2}{\sin\alpha} \qquad \sin\alpha=\frac{2}{x}\right)$$

$$\alpha=\arcsin\frac{x}{2}$$

↓ α yi x'e geri

---

67) $$\int \frac{x^3-x^2+2x-5}{5(x-1)^2(x^2+1)} = ?$$

$$\frac{x^3-x^2+2x+5}{5(x-1)^2(x^2+1)} = \frac{A}{(x-1)}+\frac{B}{(x-1)^2}+\frac{Cx+D}{(x^2+1)}$$

↳

68) $$I=\int e^{\arccos x}\,dx = ?$$

$$\left(\begin{array}{l}\arccos x = u\\ x=\cos u\\ dx=-\sin u\,du\end{array}\right) \rightarrow -\int e^u\cdot\sin u\,du$$

Laplace?

$$\left(\begin{array}{ll}\sin u = f & dv=e^u du\\ \cos u\,du = df & v=e^u\end{array}\right)$$

$$I=-\sin u\cdot e^u+\int e^u\cdot\cos u\,du$$

Laplace ?

$$\downarrow$$

$$\begin{pmatrix} \cos v = t_1 & dv_1 = e^v dv \\ -\sin v\, dv = dt_1 & v_1 = e^v \end{pmatrix}$$

$$\downarrow$$

$$\cos v \cdot e^v + \int e^v \cdot \sin v\, dv$$

$$I = -\int e^v \sin v\, dv = -\sin v \cdot e^v + \cos v \cdot e^v + \int e^v \cdot \sin v\, dv$$

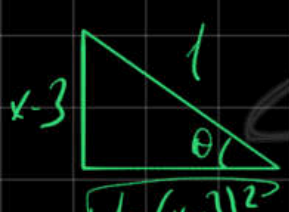
$$-2\int e^v \sin v\, dv = -\sin v\, e^v + \cos v \cdot e^v$$

$$I = +\int e^v \sin v\, dv = + \frac{\sin v \cdot e^v + \cos v \cdot e^v}{2} = \frac{e^{\arccos x}(\sin(\arccos x) + \cos(\arccos x))}{2} + c$$

74) $\int \sqrt{6x - x^2 - 8}\, dx = I = ?$

$$I = \int \sqrt{6x - x^2 - 8}\, dx = \int \sqrt{-(x^2 - 6x + 8 + 1 - 1)}\, dx = \int \sqrt{-(x^2 - 6x + 9 - 1)}\, dx$$

$$I = \int \sqrt{1 - (x-3)^2}\, dx$$



$$\sin\theta = \frac{x-3}{1} \qquad \tan\theta$$

$$\cos\theta = \frac{\sqrt{1-(x-3)^2}}{1} \qquad \cot\theta$$

$$1 = s^2 + c^2$$

$$\begin{pmatrix} \sin\theta = x - 3 \\ \sin\theta + 3 = x \\ \cos\theta\, d\theta = dx \end{pmatrix} \rightarrow I = \int \sqrt{1 - (\sin\theta)^2} \cdot \cos\theta\, d\theta = \int \cos^2\theta\, d\theta$$

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

$$I = \int \left(\frac{1+\cos 2\theta}{2}\right) d\theta = \frac{1}{2}\int 1+\cos 2\theta \, d\theta = \frac{1}{2}\left(\theta + \frac{\sin 2\theta}{2}\right) + c$$

$\sin 2\theta = 2\sin\theta.\cos\theta$
$= 2.(x-3).\sqrt{1-(x-3)^2}$ $\quad (\sin\theta = x-3,\ \theta = \arcsin(x-3))$

$$= \frac{1}{2}\left(\arcsin(x-3) + (x-3)\sqrt{1-(x-3)^2}\right) + c$$

75) $\int \frac{dx}{2\sin x + \sin 2x} = I = ?$ → Yarım açı

$$I = \int \frac{dx}{2\sin x + 2\sin x\cos x} = \int \frac{dx}{2\sin x(1+\cos x)}$$

$(\cos x = u,\ -\sin x\,dx = du)$ → $x\div$

$$\int \frac{dx(\sin x)}{2\sin^2 x(1+\cos x)} = \int \frac{dx(\sin x)}{2(1-\cos^2)(1+\cos x)}$$

$$= \int \frac{\sin x\,dx}{2(1-\cos x)(1+\cos x)^2} = -\frac{1}{2}\int \frac{du}{(1-u)(1+u)^2} = I$$

Kesirler

$$\frac{1}{(1-u)(1+u)^2} = \frac{A}{(1-u)} + \frac{B}{(1+u)} + \frac{C}{(1+u)^2}$$

$A(1+u)^2$ $\quad B(1-u)(1+u)$ $\quad C(1-u)$

$$A = \frac{1}{8},\ B = \frac{-1}{8},\ C = \frac{-1}{4}$$

$$I = -\frac{1}{2}\int \frac{\frac{1}{8}}{(1-u)} + \frac{\frac{-1}{8}}{(1+u)} + \frac{\frac{-1}{4}}{(1+u)^2} = \frac{-1}{2}\left[\frac{1}{8}\ln(1-u) - \frac{1}{8}\ln(1+u) - \frac{1}{4}(\ln($$

$u = \cos x$

76) $\int \sin^3 x \cdot \cos^{-4/3} x \, dx = I = ?$

$$I = \int \sin^2 x \cdot \sin x \cdot \cos^{-4/3} x \, dx = \int (1 - \cos^2 x) \cdot \cos^{-4/3} x \cdot \sin x \, dx$$

$$\left(\begin{array}{l} \cos x = u \\ -\sin x \, dx = du \\ \sin x \, dx = -du \end{array} \right)$$

$$= -\int (1 - u^2) u^{-4/3} \, du = -\int (u^{-4/3} - u^{2/3}) \, du = -\left(\frac{-3}{u^{1/3}} - \frac{3}{5} u^{5/3} \right) + C$$

$u = \cos x$

82) $\int \frac{(t+3)^{100}}{(t+2)^{102}} \, dt = I = ?$

$$I = \int \frac{(t+3)^{100}}{(t+2)^{100} (t+2)^2} \, dt = \int \left(\frac{t+3}{t+2} \right)^{100} \cdot \frac{1}{(t+2)^2} \, dt$$

$$\frac{t+3}{t+2} = u$$

$$\frac{1 \cdot (t+2) - (1)(t+3)}{(t+2)^2} \, dx = du \rightarrow \frac{-1}{(t+2)^2} \, dx = du$$

$$I = \int u^{100} \cdot -du = -\int u^{100}\,du = -\frac{u^{101}}{101} + C = \frac{-1}{101}\cdot\left(\frac{t+3}{t+2}\right)^{101} + C$$

84) $\int \frac{e^{2x}-1}{e^{2x}+1}\,dx = I = ?$

$x\div$

$$I: \int \frac{(e^{2x}-1)e^{-x}}{(e^{2x}+1)e^{-x}}\,dx = \int\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)dx$$

pay paydaşın türevi

$$\left(\begin{array}{l} e^x + e^{-x} = u \\ (e^x - e^{-x})\,dx = du \end{array}\right) \rightarrow \int \frac{1}{u}\cdot du = \ln|u| + C$$

$$I = \ln|e^x + e^{-x}| + C$$

91) $\int \frac{\sin(x-y)}{\sin(x+y)}\,dx = I = ?$

$$\left(\begin{array}{l} \sin(x-y) = \sin x\cos y - \sin y\cos x \\ \cos(x+y) = \cos x\cos y - \sin x\sin y \end{array}\right)$$

Tarih / Date : ...../...../.....

→ $\sqrt[3]{124}$ sayısının **Doğrusal değerini** hesaplayınız

$f(x+\Delta x) \cong f(x) + \Delta x \cdot f'(x)$

$f(x) = \sqrt[3]{x} = x^{1/3}$

$f'(x) = \frac{1}{3x^{2/3}}$

$f'(125) = \frac{1}{3 \cdot 25} = \frac{1}{75}$, $f(125) = 5$

$y - 5 = \frac{1}{75}(x - 125)$

$y = \frac{(x-125)}{75} + 5$

$y(124) = \frac{-1}{75} + 5 = \frac{374}{75}$

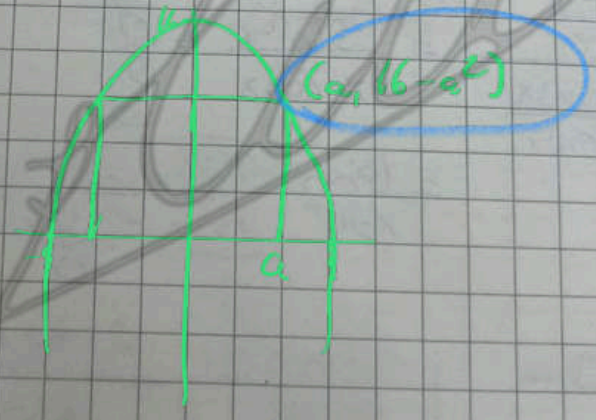
→ $y = 16 - x^2$ parabolü ile $y = 0$ doğrusu arasında kalan bölge içine çizilebilecek en büyük alanlı dikdörtgenin alanını ve boyutlarını bulunuz

$A = \frac{256}{3\sqrt{3}}$

$y = 16 - x^2$

$y = 0 \rightarrow x = 4, -4$

$x = 0 \rightarrow y = 16$



$A(a) = 2a(16 - a^2) = 32a - 2a^3$

$A'(a) = 32 - 6a^2$

$32 = 6a^2$

$16 = 3a^2$

$\frac{4}{\sqrt{3}} = a$

$A\left(\frac{4}{\sqrt{3}}\right) = \frac{256}{3\sqrt{3}}$

Tarih / Date : __/__/____

$\lim_{x \to 0^+} \left(\frac{1}{x}\right)^{\sin x} = \infty^0$

$\lim_{x \to 0^+} \left(\frac{1}{0^+}\right)^{\sin 0^+} = \infty^0$

$y = \left(\frac{1}{x}\right)^{\sin x} \qquad \ln y = \sin x \cdot \ln\left(\frac{1}{x}\right)$

$\lim_{x \to 0^+} \sin x \cdot \ln\left(\frac{1}{x}\right) = 0 \cdot \infty \rightarrow \lim_{x \to 0^+} \frac{\ln\left(\frac{1}{x}\right)}{\frac{1}{\sin x}} = \frac{\infty}{\infty}$

$\lim_{x \to 0^+} \frac{\frac{+1}{x}}{+\csc x \cdot \cot x} = \lim_{x \to 0^+} \left(\frac{\sin^2 x}{x \cdot \cos x}\right) = \frac{0}{0}$

$\lim_{x \to 0^+} \left(\frac{2 \sin x \cdot \cos x}{1 \cos x + (-\sin x) x}\right) = \frac{0}{1} = 0$

$\ln y = 0 \Rightarrow \ln_e y = 0$

$e^0 = y = 1$

Laplace

$u \cdot v - \int v \cdot du$

$\rightarrow \int x^3 \ln x \, dx$

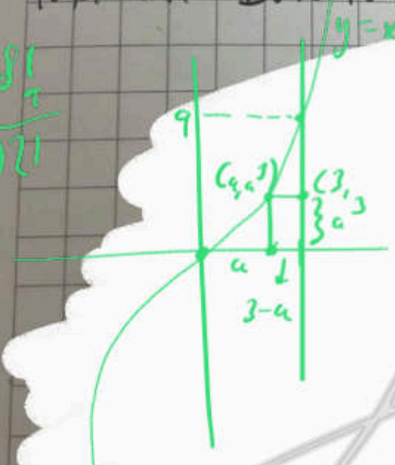
$\ln x = u \qquad x^3 dx = dv$

$\frac{1}{x} dx = du \qquad \frac{x^4}{4} = v$

$= \ln x \cdot \frac{x^4}{4} - \int \frac{x^4}{4} \cdot \frac{1}{x} dx$

$= \ln x \cdot \frac{x^4}{4} - \left(\frac{x^4}{16}\right) + C$

$\rightarrow$ $y = x^3$ eğrisi $x = 3$ doğrusu ve x-ekseni tarafından sınırlanan bölgede bulunan, bir kösesi eğri üzerinde bir köşesi doğru üzerinde ve diğer köşeside x eksen üzerinde olan maks. alanlı dikdörtgenin kenar uzunluklarını bulunuz?



$A(a) = a^2 \cdot (3-a) = 3a^3 - a^4$

$A'(a) = 9a^2 - 4a^3 = a^2(9-4a) = 0$ için

$a_1, a_2 = 0 \qquad a_3 = \frac{9}{4}$

$\left(\frac{9}{4}\right)^3 = \frac{729}{64}$

$\frac{12-9}{4} = \frac{3}{4}$

Tarih / Date : ...../...../.....

$y' = -\frac{\frac{df}{dx}}{\frac{df}{dy}}$

→ $x^3y^3 + y^2 = x + y$ (1,1) noktasındaki teğet denklemi bulunuz

$x^3y^3 + y^2 - x - y = 0$

$\frac{3x^2y^3 - 1}{3y^2x^3 + 2y - 1}$ $\quad y'(1,1) = -\frac{2}{4} = -\frac{1}{2}$

$-\frac{1}{2}$, (1,1)

$y - 1 = -\frac{1}{2}(x-1) = \frac{1-x}{2}$ $\quad y = \frac{1-x}{2} + 1$

23-24

⇒ $x = -2\cos t$
$y = 3\sin 2t$
$\Big\}\ t = \frac{\pi}{4}$ teğet denklemi?

$\frac{dx}{dt} = 2\sin t$ $\quad \frac{dy}{dt} = 6\cos 2t$ $\quad \frac{dy}{dt}\cdot\frac{dt}{dx} = y' = \frac{6\cos 2t}{2\sin t}$

$y'(\frac{\pi}{4}) = \frac{6\cdot\cos\frac{\pi}{2}}{2\cdot\sin\frac{\pi}{4}} = \frac{0}{\sqrt{2}} = 0$ $\quad t = \frac{\pi}{4} \rightarrow x = \sqrt{2}$

$y = 3$

$y - 3 = 0 \rightarrow y = 3$

2. mertebeden türev $t = \frac{\pi}{4}$ = ?

$y'' = \frac{dy'}{dx} = \frac{dy'/dt}{dx/dt}$ → $\frac{dy'}{dt} = 12\cdot -\sin 2t$

$\frac{dx}{dt} = 2\sin t$

$= \frac{6\cdot -\sin 2t}{\sin t} = \frac{-12\sin t\cos t}{\sin t}$

$y'' = -12\cos t$

$y''(\frac{\pi}{4}) = -12\cos\frac{\pi}{4} = -6\sqrt{2}$

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$\Rightarrow \int \frac{dx}{x^3+4x}$

$$\int \frac{1}{x^3+4x}\,dx = \int \frac{dx}{x(x^2+4)}$$

$$\frac{1}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$1 = Ax^2+4A+Bx^2+Cx$$

$A+B=0 \quad A=\frac{1}{4} \quad B=-\frac{1}{4} \quad C=0$

$C=0$

$4A=1$

$$\int \frac{\frac{1}{4}}{x} + \frac{\frac{-x}{4}}{x^2+4}\,dx = \frac{\ln|x|}{4} - \int \frac{x}{4x^2+16}\,dx$$

$\left(\begin{array}{l} x^2=u \\ 2x\,dx=du \\ x\,dx=\frac{du}{2} \end{array}\right) \rightarrow \frac{1}{2}\int \frac{1\,dx}{4u+16} = \frac{1}{2}\ln|4u+16|$

$= \frac{\ln|4x^2+16|}{2}$

$= \frac{\ln|x|}{4} - \frac{\ln|4x^2+16|}{2}$

$$\ln|u| = \ln|1+\sin x| + c$$

$$\Rightarrow \frac{4x}{x^2-4}$$

① $f_t = R - \{-2, +2\}$

② $x=0 \;.\; y=0$

③ $f'(x) = \dfrac{4(x^2-4)-(2x)(4x)}{(x^2-4)^2} = \dfrac{-4(x^2+4)}{(x^2-4)^2}$

$4 \; x_1 = ?$
$x_2 = -2$

④ Kontrol

⑤ $x=2$ $\quad x=-2$ dikey asy

$\lim_{x\to 2^+} \frac{4x}{x^2-4} = \frac{8}{0} = \text{sonsuza gidiyor} \to \frac{+}{+} = +\infty \qquad \lim_{x\to 2^-} \frac{4x}{x^2-4} = \frac{+}{-} = -\infty$

$\lim_{x\to -2^+} \frac{4x}{x^2-4} = \frac{-8}{0} \to \frac{-}{+} = +\infty \qquad \lim_{x\to -2^-} \frac{4x}{x^2-4} \to \frac{-}{+} = -\infty$

$\lim_{x\to\infty} \frac{4x}{x^2-4} = \frac{x(4)}{x^2(1-\frac{4}{x^2})} = \frac{4}{x} = 0$ $\quad (y=0)$ yatay asy

$\lim_{x\to-\infty} f(x) = 0$

⑥

		-2		2	
f'	−	‖	+	‖	−
	↘	‖	↗	‖	↘
	0	$-\infty$ ∞		$-\infty$ ∞	0

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$\Rightarrow y = \frac{x}{x^2-1}$ = grafiğini çizmiş.

① $\mathbb{R} - \{-1, 1\}$

② $x = 0 \rightarrow y = 0$

③ $f'(x) = \frac{1(x^2-1) - (2x)(x)}{(x^2-1)^2} = \frac{-x^2-1}{(x^2-1)^2} \rightarrow \phi$

$x_{1,2} = 1$
$x_{3,4} = -1$ Gitti

④ Kırılım

⑤ $f(x) = \frac{x}{x^2-1}$ $x = 1$ $x = -1$ düşey asp

$\lim_{x \to 1^+} \frac{x}{x^2-1} = \frac{1}{0} \rightarrow \frac{+}{+} = +\infty$ $\lim_{x \to 1^-} \frac{x}{x^2-1} = \frac{+}{-} = -\infty$

$\lim_{x \to -1^+} \frac{x}{x^2-1} = \frac{-}{-} = +\infty$ $\lim_{x \to -1^-} \frac{x}{x^2-1} = \frac{-}{+} = -\infty$

$\lim_{x \to \infty} \frac{x}{x^2-1} = \frac{\infty}{\infty} \rightarrow \lim_{x \to \infty} \frac{1}{2x} = 0$

$\lim_{x \to -\infty} \frac{x}{x^2-1} = \frac{-\infty}{\infty} \rightarrow \lim_{x \to -\infty} \frac{1}{2x} = 0$

$y = 0$ yatay asp

⑥

x		-1		1	
$f'(x)$	$-$	$\nexists$	$-$	$\nexists$	$-$
$f(x)$	0 $\searrow$	$-\infty \mid\mid \infty$	$\searrow$	$-\infty \mid\mid \infty$	$\searrow$ 0

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